Teaching notes and solutions

Teacher notes

This chapter is really finished at the beginning of May, not at the end of April. It therefore does not feature in April's second mock exam. Revisit the circle before the curves. Use radians in derivatives. Have students check the endpoints $-\pi$ and π: they represent the same point of the circle but are two distinct numbers if the interval contains them. The variations of composites are deduced from the full sign of the derivative.

Solutions to the activity

1. $h(0) = 3 - 2 = 1$ m. Adding 8 to t adds 2π to the argument of the cosine: 8 s is a period, and it is the smallest positive period. Since $-1 \leq \cos \leq 1$, $1 \leq h \leq 5$; both bounds are attained.

2. By differentiating a composite, $h'(t) = (\frac{\pi}{2})\sin(\frac{\pi t}{4})$. The argument ranges over $[0, 2\pi]$. Sine is positive on $]0, \pi[$, then negative on $]\pi, 2\pi[$. Thus h increases from 1 to 5 m on $[0, 4]$, then decreases from 5 to 1 m on $[4, 8]$.

3. $h(t) = 4 \iff \cos(\frac{\pi t}{4}) = -\frac{1}{2}$. On $[0, 2\pi]$, the corresponding angles are $\frac{2\pi}{3}$ and $\frac{4\pi}{3}$; so $t = \frac{8}{3}$ s or $t = \frac{16}{3}$ s.

4. The condition $h(t) \geq 4$ is equivalent to $\cos(\frac{\pi t}{4}) \leq -\frac{1}{2}$, hence to $t \in [\frac{8}{3}, \frac{16}{3}]$. The duration is $\frac{16}{3} - \frac{8}{3} = \frac{8}{3}$ s.

E1. On the circle, sine equals $-\frac{1}{2}$ at the angles $-\frac{\pi}{6}$ and $-\frac{5\pi}{6}$ in $[-\pi, \pi]$. The other angles are deduced by adding $2k\pi$ and lie outside the required interval.

E2. Cosine is at least $\frac{\sqrt{2}}{2}$ on the arc around the angle 0, between $-\frac{\pi}{4}$ and $\frac{\pi}{4}$, endpoints included. The solution set is therefore $[-\frac{\pi}{4}, \frac{\pi}{4}]$.

E3. By composition, $(\cos(3x - 1))' = -3\sin(3x - 1)$. For the square of sine, $(\sin^2 x)' = 2\sin x \cos x$. Both derivatives are defined on $\mathbb{R}$.

E4. $f'(x) = \cos x$, positive on $[0, \frac{\pi}{2}[$ and $]\frac{3\pi}{2}, 2\pi]$, negative on $]\frac{\pi}{2}, \frac{3\pi}{2}[$. Thus f increases from 2 to 3, decreases from 3 to 1, then increases from 1 to 2. The maximum is 3 at $\frac{\pi}{2}$ and the minimum 1 at $\frac{3\pi}{2}$.

E5. An antiderivative of sine is $-\cos x$. So the integral equals $[-\cos x]_0^{\pi} = 1 - (-1) = 2$. The mean value over this interval of length π is $\frac{2}{\pi}$.

Solutions to Homework 5

1. By the product rule, $f' = \cos^2 x - \sin^2 x = \cos 2x$. On the interval studied, $2x \in [0, \pi]$: the derivative is positive before $x = \frac{\pi}{4}$ and negative after. The maximum is $f(\frac{\pi}{4}) = (\frac{\sqrt{2}}{2})^2 = \frac{1}{2}$.

2. Thanks to $\sin 2x = 2\sin x \cos x$, $f(x) \geq \frac{1}{4}$ is equivalent to $\sin 2x \geq \frac{1}{2}$. For $2x \in [0, \pi]$, this gives $2x \in [\frac{\pi}{6}, \frac{5\pi}{6}]$, so $x \in [\frac{\pi}{12}, \frac{5\pi}{12}]$.

3. The derivative of $\frac{1}{2}\sin^2 x$ is $\sin x \cos x = f(x)$. So $\int_0^{\frac{\pi}{2}} f(x)\,\mathrm{d}x = [\frac{1}{2}\sin^2 x]_0^{\frac{\pi}{2}} = \frac{1}{2}$. The mean value is $\frac{\frac{1}{2}}{\frac{\pi}{2}} = \frac{1}{\pi}$.

4. $A(a) = \frac{1}{2}\sin^2 a$. Half of the total area is $\frac{1}{4}$, so $\sin^2 a = \frac{1}{2}$. On $[0, \frac{\pi}{2}]$, sine is positive and strictly increasing: this equality is equivalent to $\sin a = \frac{\sqrt{2}}{2}$, then to $a = \frac{\pi}{4}$.

Solutions to the textbook or past-paper work

The curve of $f(x) = \cos(3x)$ suggests a symmetry about the vertical axis and a repetition after a horizontal translation of $\frac{2\pi}{3}$. For every $x \in \mathbb{R}$, $f(-x) = \cos(-3x) = \cos(3x) = f(x)$: f is even. Moreover,

$$f\left(x + \frac{2\pi}{3}\right) = \cos(3x + 2\pi) = \cos(3x) = f(x).$$

The number $\frac{2\pi}{3} > 0$ is therefore a period of f. The drawing suggests these properties; the identities establish them for every real number.